

UniGPT

An AI Academic Copilot for Smart University Management

Team Information

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1. Problem Statement

Modern universities still rely heavily on fragmented systems and manual workflows for managing academic activities. Students frequently struggle to access important course materials, notices, assignment guidelines, attendance records, and academic policies because these resources are scattered across multiple disconnected platforms such as ERP systems, WhatsApp or Messenger groups, Google Classroom, spreadsheets, and paper-based records.

Faculty members face similar challenges, spending significant time on repetitive tasks such as attendance tracking, assessment creation, grading, feedback generation, and academic communication. Administrative departments also lack real-time institutional visibility, making decision-making slow and inefficient.

2. Proposed Solution

UniGPT is an intelligent, centralized academic platform powered by Artificial Intelligence. Rather than forcing students, faculty, and administrators to navigate multiple disconnected tools, UniGPT unifies all major academic workflows into a single platform with role-based experiences. The system provides one integrated ecosystem for academic communication, course management, attendance, assessments, document sharing, analytics, and AI-assisted decision support.

The entire university is organised around a clear academic structure — departments contain courses, courses are offered as class sections each term, and students enrol into those sections. Everything else (materials, assignments, quizzes, attendance, exams, and grades) belongs to the specific section a student is actually in, so each user only ever sees content relevant to their own classes.

The core innovation behind UniGPT is its Retrieval-Augmented Generation (RAG) architecture. Unlike generic AI chatbots that may produce inaccurate or hallucinated responses, UniGPT grounds every answer in institution-approved documents — syllabi, policies, handbooks, and lecture materials. Every

response includes source citations (down to the document and page) and confidence scores, making the system trustworthy and auditable.

Dual-mode AI engine: UniGPT runs in either a Live mode powered by OpenAI for the smartest answers, or a built-in offline Demo mode that needs no API key at all — so the entire product, including AI chat and AI generation, can be demonstrated with zero external setup or cost. If Live mode is selected but no key is configured, the system automatically falls back to Demo mode rather than failing.

3. Objectives

- Provide students, faculty, and administrators with a single unified platform for all academic activities, eliminating the need to switch between multiple disconnected tools.
- Deliver accurate, hallucination-free AI responses by grounding every answer in verified institutional documents through RAG architecture, with citations and confidence scores.
- Reduce repetitive administrative and faculty workload by automating tasks such as quiz generation, attendance tracking, grading feedback, and academic announcements.
- Enable real-time institutional visibility for administrators through centralized analytics, AI-usage monitoring, early risk detection, and document management.
- Build a permission-aware, role-based AI system where each user — student, faculty, or admin — only accesses information relevant and authorized to their role.
- Guarantee the system runs anywhere by offering a fully offline AI demo mode alongside the live AI provider, removing any hard dependency on external services.

4. Methodology / Technology Used

UniGPT is built on a modern full-stack architecture and an AI pipeline designed for reliability and scale:

- **RAG Pipeline:** User queries and institution-approved documents are converted into vector embeddings. A semantic similarity search retrieves the most relevant document chunks, which are then passed as context to the language model for grounded response generation. The relevance search respects each user's permissions, so two people asking the same question may receive answers built from different sources.
- **Backend:** Laravel 11 (PHP) provides secure REST APIs, role-based access control, and a modular service architecture.
- **Frontend:** Vue.js 3 with Inertia.js and Tailwind CSS delivers a fast, responsive, and role-differentiated user interface.
- **Database & Vector Search:** PostgreSQL handles structured academic data (users, courses, attendance, assessments, document metadata) and also stores document embeddings for semantic retrieval — so vector search runs inside the existing database, with no separate specialised search engine to operate.
- **AI Models:** OpenAI GPT-4o for reasoning and response generation, and text-embedding-3-large for semantic document retrieval, in Live mode. All AI parameters — provider, models, creativity, number of source passages per question, minimum relevance threshold, tone, and supported answer languages — are configurable by admins.

- **Dual AI Modes:** An admin-switchable Live mode (OpenAI) and an offline Demo mode that requires no API key, with automatic fallback, so the platform is fully demonstrable with zero external cost.
- **AI Chat Modes:** Students can switch between academic support, exam preparation, research assistance, assignment help, and career guidance modes, and choose their preferred answer language.

5. Expected Outcomes

Once deployed, UniGPT is expected to deliver the following measurable improvements:

- Significant reduction in the time students spend searching for academic information, as all course materials, policies, and notices become instantly accessible through a single AI-powered interface.
- Decreased faculty workload on repetitive tasks such as quiz creation, grading, and attendance management, allowing more time for quality teaching.
- Higher accuracy and trustworthiness of academic information delivery, as AI responses are grounded in approved documents rather than generic internet data.
- Improved administrative decision-making through real-time dashboards, analytics, AI-usage monitoring, and early risk detection for students showing engagement or performance decline.
- A measurable reduction in academic miscommunication, as all official notices, policies, and announcements flow through one verified, centralized platform.

6. Core Features

UniGPT offers comprehensive features tailored to three major user groups:

- **Students:** AI academic copilot with grounded, cited Q&A; multiple chat modes (academic support, exam prep, research, assignment help, career guidance); saved answers; degree roadmap and transcript; course registration; attendance tracking; GPA/CGPA monitoring; a combined calendar; and personal notes and tasks for deadline management.
- **Faculty:** AI teaching assistant for generating quizzes, assignments, and grading feedback; managing attendance; publishing materials; auto-grading proctored class tests; submitting documents for approval; and analytics that flag at-risk students.
- **Administrators:** Centralized control over users, departments, terms, courses and sections, the document library and approval workflow, exams, announcements, institution-wide analytics, and the AI settings.

Platform-wide capabilities:

- **Document approval workflow:** every uploaded document moves through a review queue where admins can approve, reject (with reason), request changes, or comment. Approval automatically reads, splits, and indexes it into the AI's knowledge; rejecting or editing it removes it from AI answers instantly.
- **Proctored quizzes:** class tests run under fullscreen proctoring with a tamper-proof timer; correct answers never reach the browser during a live attempt, tab-switching is recorded as a warning, and excessive violations can disqualify an attempt. Submissions are graded instantly.

- **Granular access control:** a visual role-and-permission editor, fine-grained permissions, temporary roles that auto-expire, and a protected admin account that can never be locked out.
- **AI usage monitoring:** admins can track per-user AI usage and revoke or reinstate an individual's AI-chat access to control cost or misuse.
- **Messaging & system health:** students and their own teachers can message directly (only when they share a class), with live presence indicators, plus a live system-health monitor for database, storage, jobs, and AI status.

7. Impact and Feasibility

From a technical perspective, UniGPT is highly feasible and production-ready. The system has already implemented major admin, faculty, and student functionalities — including the dual-mode AI engine, document approval and indexing, proctored assessments, and messaging — using proven, scalable technologies. Its modular architecture supports secure role-based access control, efficient in-database document indexing, and maintainable long-term development.

The societal impact of UniGPT is significant. In a country like Bangladesh where universities often struggle with resource fragmentation and inconsistent academic communication, a centralized AI-powered platform can meaningfully raise the quality and equity of education. By making institutional knowledge instantly accessible to every student and reducing administrative friction for faculty, UniGPT has the potential to improve academic outcomes across institutions of all sizes.

Technologically, UniGPT demonstrates how RAG-based AI can be responsibly deployed in high-stakes academic environments — setting a model for trustworthy, citation-backed AI usage that other institutions can follow.

8. Scalability & Future Vision

UniGPT is designed with scalability in mind. Its architecture can support thousands of students, faculty members, departments, and documents across multiple institutions, with an end-of-term rollover process that winds classes down, locks in grades, and promotes the next semester automatically.

Direct messaging and presence are already built into the platform. Future expansion includes integration with external platforms like WhatsApp and Telegram, predictive analytics, voice-based AI interaction, digital library integration, and broader support for additional AI providers beyond the current configurable OpenAI and offline demo backends.

The long-term vision is to transform universities into AI-driven intelligent ecosystems where academic knowledge becomes instantly accessible, administrative friction is minimized, and educational productivity is significantly enhanced.